

MES Current-State Assessment & Target-State Blueprint

Formal assessment report for fictional regulated life sciences manufacturing

Disclaimer and use

Portfolio artifact using fictional, non-proprietary examples. Not based on confidential employer data.

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Executive Summary

This report presents a fictional current-state assessment and target-state blueprint for MES-enabled manufacturing. The assessment focuses on execution variability, fragmented SOP access, disconnected QC/LIMS signals, manual handoffs, delayed QA readiness, and limited operational visibility.

Objective and scope

- Assess process friction across Manufacturing, QA, QC/LIMS, Scheduling, MES Engineering, and Automation/IT-OT.
- Frame a future-state operating model across ERP, MES, Automation/DeltaV, LIMS/QC, Process Data/OSI PI, and QA Review.
- Identify practical recommendations and success measures for a regulated digital manufacturing program.

Assessment lens

- How work is triggered, executed, verified, reviewed, and released.
- Where information is fragmented or manually reconciled.
- Which handoffs create readiness delays or avoidable support demand.

Current-State Observations by Function

Manufacturing	Execution prompts and reference access vary by product and line. Operators rely on tribal knowledge when SOP context or exception pathways are not embedded in the flow.
QA	Review package completeness is often discovered late, creating manual reconciliation and delayed release readiness.
QC/LIMS	Sample status and result visibility are not consistently available in the batch execution context.
Scheduling	Production priorities shift faster than readiness signals can be reconciled across teams.
MES Engineering	Support queues mix defects, change requests, master data questions, and operator how-to support.
Automation / IT-OT	DeltaV/process data context exists but is not always connected to review or investigation needs.

Gap Assessment Matrix

Current-state issue	Operational impact	Target-state recommendation	Priority
Fragmented SOP access	Search time, wrong-version risk, inconsistent execution support	Step-level controlled references surfaced in MES	High
Disconnected QC/LIMS status	Release decisions require follow-up and manual status checks	Sample status and result readiness visible in batch context	High
Delayed QA readiness	Review blockers found late in the process	Review readiness dashboard with missing item and exception indicators	High
Manual handoffs	Scheduling and execution readiness drift apart	Shared readiness model across operations, QA, QC, and MES support	Medium
Limited process data linkage	Investigation context is collected manually	Process data/OSI PI signals linked to batch review context	Medium

Target-State Operating Model

Target-state system flow

ERP -> MES -> Automation / DeltaV -> LIMS / QC -> Process Data / OSI PI -> QA Review

ERP

Production demand, material context, planning signals

MES

Recipe execution, EBR workflow, verification, signatures, deviations

Automation / DeltaV

Equipment/process execution and parameter context

LIMS / QC

Sample status, results, release-impacting quality context

Process Data / OSI PI

Historian trends and investigation context

QA Review

Structured readiness, exceptions, evidence, and release decision support

Recommendations and Roadmap Considerations

Recommendations

- Standardize execution prompts, verification checkpoints, and exception routing in MES.
- Embed controlled SOP/reference access near execution steps.
- Connect QC/LIMS readiness and process data context to batch review workflows.
- Define clear ownership for master data, recipe changes, support triage, and review readiness.

Roadmap considerations

- Prioritize high-impact readiness gaps before broad configuration changes.
- Align QA, operations, automation, QC, and MES engineering on acceptance criteria.
- Pilot one representative workflow before scaling across products or lines.

Success Measures and Conclusion

Review readiness	Earlier visibility into missing items, exceptions, and QC status
Execution quality	Improved right-first-time execution and reduced correction effort
Deviation context	More complete batch, process, and sample context for triage
Schedule adherence	Improved alignment between production schedule and system readiness
Support demand	Clearer separation of defects, enhancements, training needs, and master data issues

Conclusion: A cleaner MES target state is not only a technology configuration exercise. It requires operating model clarity, controlled data flow, cross-functional ownership, and success measures that connect execution quality to review readiness and manufacturing visibility.